

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING
 Mini-Test August-2025

Course: B.Tech		Semester - III Section- II	Branch: Electrical
Subject: Electrical Machines-I		Subject Code: EE-24212	
Max Marks: 10			
Q.No.	Question	Marks	
1.	A 440 Volts, 3-phase, 4-wire system supplies an unbalanced load represented by following impedances: $Z_R= 16+j12$, $Z_Y= 14-j21$, $Z_B= 25$ ohms. The phase sequence is RYB. Calculate (i) current in each line (ii) current in neutral wire (iii) Readings on each of the three watt meters connected between line and neutral. Draw corresponding phasor diagram.	03	
2.	A 3-phase, 4-wire, 415/240 V system feeds unbalanced lighting loads on the three phases. The lamp currents are: Phase A: 96 A Phase B: 52 A Phase C: 74 A (Assume unity power factor for the lamps.) Find the neutral current with the lamps only. A 3-phase motor is then started, taking 150 A per line at 0.35 power factor (lagging). Find the current in each line and the neutral current. Find the total real power supplied to the lamps and to the motor.	04	
3.	A wattmeter reads 8 kW when its current coil is connected in R-phase and its voltage coil is connected between the neutral and R-phase of a symmetrical 3-phase system supplying a balanced load of 40 A at 400 volts. What will be the reading on the instrument if the connections to the current coil remain unchanged and the voltage coil is connected between B and Y phase? Take phase sequence RYB. Draw corresponding phasor diagram.	03	